

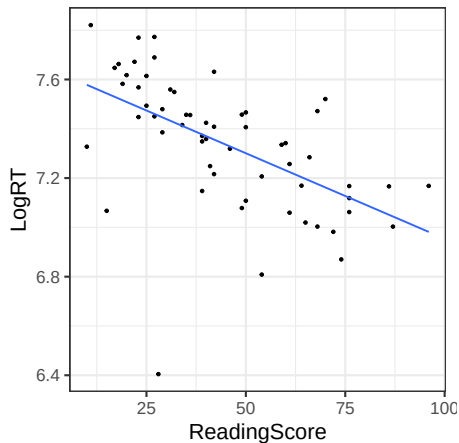
STATISTICS FOR LINGUISTS

DATA TRANSFORMATIONS

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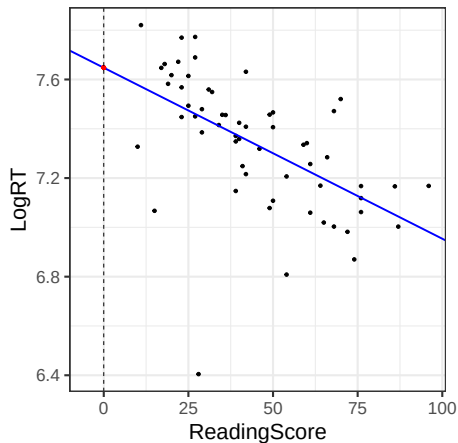
WHY BOTHER WITH STATISTICS?

Little Dutch readers



$$\text{LogRT} = 7.648 - 0.007 \times \text{ReadingScore}$$

Problems with the intercept



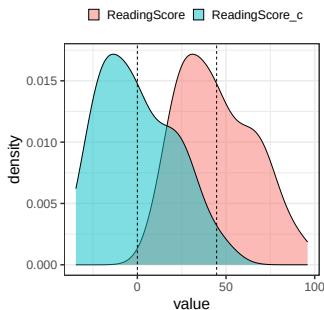
$$\text{LogRT} = 7.648 - 0.007 \times \text{ReadingScore}$$

Problems with the intercept

In this case, the intercept corresponds to the reaction time of a child with the reading score of 0, which makes very little sense.

We can certainly put the intercept to better use, for instance, we could make it correspond to the expected reaction time for a child with the mean reading score in this sample.

Adding a constant

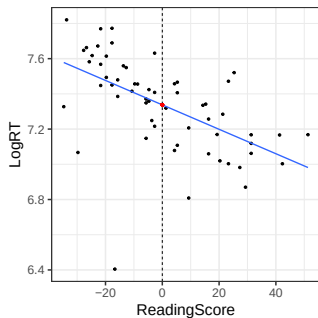


We know that adding a constant to each data point, moves the mean by the same amount.

Thus, if we subtract the sample mean from each data point, the resulting sample mean will be equal to zero.

This is known as *centering*.

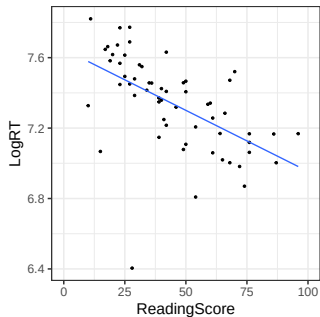
Centered predictor



The intercept now corresponds to the expected reaction time of child with a mean reading score.

$$\text{LogRT} = 7.338 - 0.007 \times \text{ReadingScore}$$

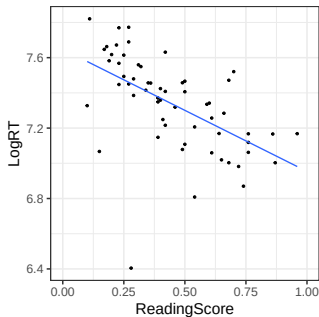
The original model again



Reading score is expressed on a scale from 0 to 100.

$$\text{LogRT} = 7.648 - 0.007 \times \text{ReadingScore}$$

Problem with the slope



Let's express it on a scale from 0 to 1 instead.

The slope will correspond to a change covering the whole scale.

However, the actual underlying effect remains unchanged!

$$\text{LogRT} = 7.648 - 0.7 \times \text{ReadingScore}$$

Problem with the slope

As we can see, the value of slope depends on the unit of the dependent variable.

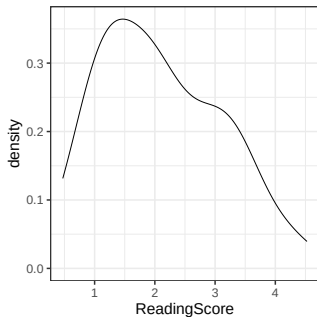
This is quite unfortunate, especially when we start adding more predictors, each expressed in different units.

Multiplying by a constant

As you remember, if we divide each data point in a sample by a constant value, the standard deviation of the sample will be reduced by the same factor.

Thus, if we divide the independent variable by its standard deviation, its standard deviation will become 1.

Standard deviation as a unit



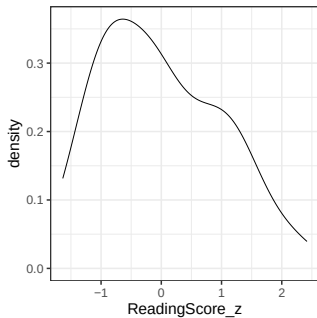
In other words, we can express the variable in units of standard deviation!

Standardizing

Usually, we combine both these transformations:

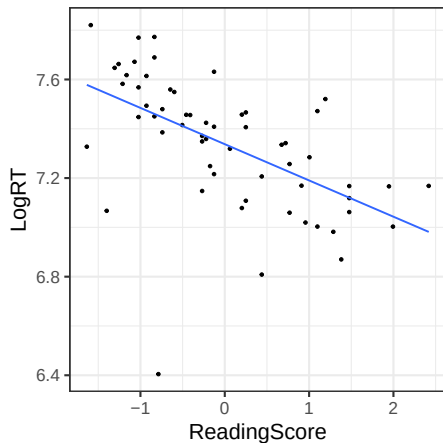
- First, we subtract the mean from each data point.
- Next, we divide each data point by the standard deviation of the sample.

Standardising



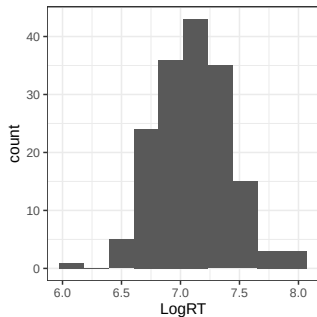
The resulting variable has a mean of 0 and standard deviation of 1.

Standardized predictor



$$\text{LogRT} = 7.338 - 0.147 \times \text{ReadingScore}$$

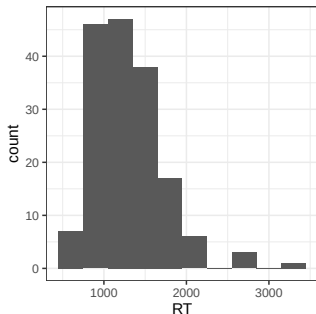
A closer look at reaction times



These reaction times look are really long!

This is because they have been expressed on a log-scale.

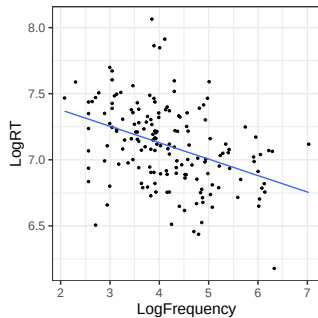
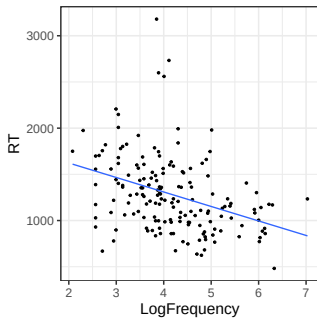
Raw reaction times



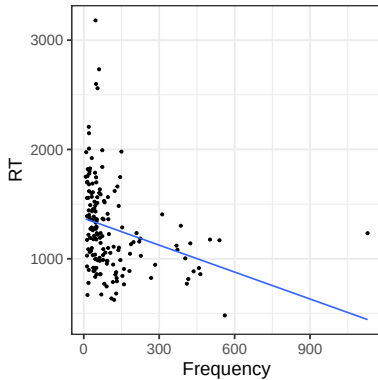
The raw reaction times have positive skew.

When the dependent variable has a distribution like this, it will often violate assumption of regression analysis.

Before and after

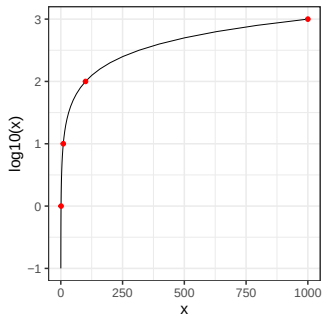


What about LogFrequency?



WHAT IS THE LOG TRANSFORMATION AND WHY DOES IT HELP?

Log scale



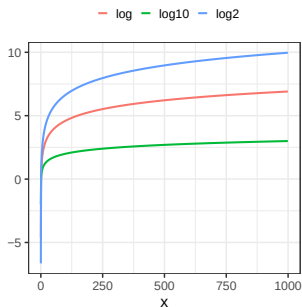
This is the $\log_{10}$ function.

As you can see, increasingly large steps on the x -axis (1, 10, 100, 1000) correspond to constant steps on the y -axis (0, 1, 2, 3,)

Log scale – $\log_{10}$

x	$\log_{10}(x)$
0.000001	-6
0.00001	-5
0.0001	-4
0.001	-3
0.01	-2
0.1	-1
1	0
10	1
100	2
1000	3
10000	4
100000	5
1000000	6

Other log-scales



Note that there are many different logarithmic functions with somewhat different properties.

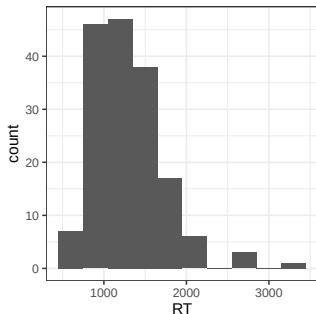
Log scale – $\log_2$

x	$\log_2(x)$
0.016	-6
0.031	-5
0.062	-4
0.125	-3
0.25	-2
0.5	-1
1	0
2	1
4	2
8	3
16	4
32	5
64	6

Log scale – natural logarithm

x	$\log(x)$
0.002	-6
0.007	-5
0.018	-4
0.050	-3
0.135	-2
0.368	-1
1	0
2.718	1
7.389	2
20.086	3
54.598	4
148.413	5
403.429	6

Log-transformation



The log scale compresses large numbers more than small numbers.

For this reason, it allows us to "fold in" the long tail and to obtain a more symmetric distribution.

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